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Dr. Kim's Fall 2026 MTH 207: HW05 (portion to submit). **Please look online at the FULL version.**
Due Thursday October 15.

1. If $f(x) = 2^x \cdot x^3$ find $f'(10)$

$f'(x) = 2^x \cdot 3x^2 + x^3 \cdot 2^x \ln 2$

$f'(10) = 2^{10} \cdot 3(10)^2 + (10)^3 \cdot 2^{10} \ln(2)$

2. If $y = \dfrac{x^4 + x^3 + x^2 + x + 1}{x^2}$ find $\dfrac{dy}{dx}$

$y = x^2 + x + 1 + x^{-1} + x^{-2}$

So $\dfrac{dy}{dx} = 2x + 1 + 0 + (-1)x^{-2} + (-2)x^{-3}$

Alternate answer

$$\frac{dy}{dx} = \frac{x^2 \cdot (4x^3 + 3x^2 + 2x + 1 + 0) - (x^4 + x^3 + x^2 + x + 1) \cdot 2x}{(x^2)^2}$$

(Parens needed due to Order of Operations)

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3. Given $f(x) = 3\sin x + 5\cos x$ find $f'(\frac{2\pi}{3})$

$f'(x) = 3\cos x + 5(-\sin x)$

$f'(\frac{2\pi}{3}) = 3\cos\frac{2\pi}{3} + 5(-\sin\frac{2\pi}{3})$

$= 3(-\frac{1}{2}) + 5(-(\frac{1}{2}))$

4. Given $f(x) = x^3 \sin x$ find $f'(x)$

$f'(x) = x^3\cos x + \sin x \cdot 3x^2$

negative slopes

horizontal tangent line

positive slopes

5. Draw the graph of a function $y = f(x)$ in which $f'(x)$ is zero when $x = 2$, and $f'(x) > 0$ if $x < 2$, and $f'(x) < 0$ when $x > 2$.

Here are examples







